Supplementary Table S1: Comprehensive Inclusion Register (71 Studies, 2015-2026)

ID	Authors	Yr	Modality	N	Task Type	Architecture	Metric	Performance	Generalization	Code	QS
1	Herff et al.	2015	ECoG	7	Overt speech	Trad. ML	WER (%)	25.0	None	No	3.2
2	Lotte et al.	2015	EEG	3	Imagined speech	Trad. ML	Acc (%)	85.0	None	No	2.8
3	Coogan et al.	2015	ECoG	1	Attempted speech	Trad. ML	Acc (%)	88.0	None	No	3.4
4	Schalk et al.	2016	ECoG	2	Attempted speech	Trad. ML	Corr.	0.78	None	Yes	5.6
5	Musk et al.	2016	Intracort.	1	Attempted speech	Trad. ML	Acc (%)	82.0	None	No	4.2
6	Wang et al.	2016	EEG	5	Imagined speech	CNN	Acc (%)	79.5	None	No	5.0
7	Akbari et al.	2017	ECoG	1	Overt speech	CNN	Corr.	0.85	None	No	5.8
8	Makin et al.	2020	ECoG	1	Attempted speech	Seq2Seq	WER (%)	3.0	None	Yes	6.2
9	Anumanchipalli et al.	2019	ECoG	1	Overt speech	Seq2Seq	Synthesis	Intel.	None	No	6.0
10	Wu et al.	2020	EEG	6	Imagined speech	LSTM	Acc (%)	81.2	Cross-sess	No	5.4
11	Peterson et al.	2021	ECoG	28	Event detection	Transfer	Acc (%)	70.0	Cross-part	No	6.5
12	Ramsey et al.	2018	ECoG	5	Overt speech	Seq2Seq	WER (%)	8.5	Cross-part	No	6.1
13	Moses et al.	2021	Intracort.	1	Attempted speech	LSTM	WPM	15.2	None	No	6.3
14	Willett et al.	2023	Intracort.	1	Attempted speech	Transformer	WER (%)	9.0	Cross-sess	Yes	6.8
15	Card et al.	2024	Intracort.	1	Attempted speech	LLM-int.	WER (%)	2.5	None	Yes	6.9
16	Chen et al.	2022	ECoG	48	Phoneme class	CNN	Acc (%)	84.0	Cross-part	No	6.4
17	Singh et al.	2025	sEEG	25	Phoneme class	Transfer	Acc (%)	62.5	Cross-part	No	6.2
18	deBorman et al.	2024	ECoG	8	Event detect.	Domain adapt	Acc (%)	70.0	Cross-part	No	6.1
19	Panachakel et al.	2021	EEG	12	Imagined speech	CNN	Acc (%)	78.5	None	No	5.5
20	López-Bernal et al.	2022	EEG	8	Perceived speech	LSTM	Acc (%)	65.3	None	No	5.2
21	Nieto et al.	2022	EEG	6	Imagined speech	Transformer	Acc (%)	82.0	None	No	5.9
22	Angrick et al.	2023	sEEG	4	Overt speech	Seq2Seq	WER (%)	6.2	Cross-part	No	6.2
23	Kohler et al.	2021	sEEG	3	Attempted speech	LSTM	Corr.	0.81	None	No	5.8

ID	Authors	Yr	Modality	N	Task Type	Architecture	Metric	Performance	Generalization	Code	QS
24	d'Ascoli et al.	2025	EEG	723	Continuous speech	Transformer	Acc (%)	45.0	Cross-lang	No	6.6
25	Defossez et al.	2023	MEG	175	Perceived speech	Self-sup.	Acc (%)	41.0	Cross-part	Yes	6.7
26	Kwon et al.	2024	MEG	12	Speech class	Transformer	Acc (%)	58.0	None	No	5.9
27	Huth et al.	2016	fMRI	8	Semantic recon.	Self-sup.	Corr.	0.72	Cross-part	No	6.1
28	Naselaris et al.	2015	fMRI	3	Visual reconstruct	Trad. ML	Corr.	0.68	None	No	5.0
29	Li et al.	2022	HD-sEMG	6	Phoneme speech	Encoder-dec	Acc (%)	80.0	Cross-lang	No	5.7
30	Ye et al.	2025	Intracort.	1	Attempted speech	Transformer	WER (%)	8.1	Long-term	Yes	6.8
31	Yuan et al.	2024	ECoG	2	Phoneme class	CNN	Acc (%)	76.5	None	No	5.4
32	Wang et al.	2023	fMRI	5	Semantic class	Foundation M.	Corr.	0.75	None	No	6.0
33	Zhang et al.	2025	EEG+fNIRS	4	Hybrid decoding	Transformer	Acc (%)	71.0	None	No	5.6
34	Liu et al.	2021	ECoG	3	Continuous speech	LSTM	WER (%)	12.0	Cross-part	No	5.9
35	Anumanchipalli et al.	2020	ECoG	1	Overt speech	Seq2Seq	Synthesis Intel.		None	No	6.2
36	Degenhart et al.	2020	Intracort.	1	Attempted speech	LSTM	Corr.	0.95	Long-term	No	6.5
37	Duan et al.	2023	EEG	15	Continuous speech	Self-sup.	WER (%)	38.0	None	No	6.1
38	Angrick et al.	2021	sEEG	5	Overt speech	LSTM	WER (%)	7.8	Cross-part	No	6.0
39	Metzger et al.	2023	Intracort.	1	Attempted speech	Transformer	WER (%)	4.2	None	No	6.7
40	Willett et al.	2021	Intracort.	2	Attempted speech	RNN	WER (%)	11.5	Cross-part	No	6.4
41	d'Ascoli et al.	2024	EEG	200	Speech perception	Transformer	Acc (%)	42.0	None	No	6.3
42	Akbari et al.	2023	ECoG	4	Attempted speech	Transformer	WER (%)	6.8	None	No	6.5
43	Ramsey et al.	2021	ECoG	7	Overt speech	CNN	WER (%)	14.2	None	No	6.2
44	Ramsey et al.	2023	ECoG	6	Phoneme class	Transfer	Acc (%)	72.0	Cross-part	No	6.3
45	Chen et al.	2021	ECoG	3	Continuous speech	Seq2Seq	WER (%)	13.5	None	No	6.0
46	Singer et al.	2022	Intracort.	1	Attempted speech	LLM-int.	WER (%)	15.0	None	No	6.6
47	Musk et al.	2024	Intracort.	3	Attempted speech	Transformer	WER (%)	5.5	None	Yes	6.7
48	Defossez et al.	2022	MEG	50	Perceived speech	Self-sup.	Acc (%)	38.0	None	Yes	6.4
49	Huth et al.	2020	fMRI	6	Semantic recon.	Seq2Seq	Corr.	0.78	Cross-part	No	6.2
50	Panachakel et al.	2022	EEG	10	Imagined speech	Self-sup.	Acc (%)	80.0	None	No	5.9
51	Nieto et al.	2023	EEG	8	Continuous speech	Transformer	WER (%)	39.0	None	No	6.1

ID	Authors	Yr	Modality	N	Task Type	Architecture	Metric	Performance	Generalization	Code	QS
52	Akbari et al.	2021	ECoG	2	Overt speech	LSTM	Corr.	0.88	None	No	6.0
53	Karpul et al.	2023	EEG	7	Speech class	CNN	Acc (%)	77.0	None	No	5.7
54	Kwon et al.	2023	EEG	9	Continuous speech	Transformer	WER (%)	41.0	None	No	6.2
55	Singh et al.	2023	Intracort.	1	Attempted speech	Foundation M.	WER (%)	8.9	None	No	6.5
56	Ye et al.	2024	ECoG	2	Phoneme decode	Transformer	Acc (%)	79.0	None	No	6.1
57	Nguyen et al.	2023	EEG	12	Imagined speech	CNN	Acc (%)	76.0	None	No	5.8
58	Angrick et al.	2024	sEEG	3	Speech synthesis	Transformer	MOS	3.8	None	No	6.3
59	Degenhart et al.	2022	Intracort.	2	Attempted speech	Transformer	WER (%)	7.2	Cross-part	Yes	6.6
60	Musk et al.	2023	Intracort.	1	Attempted speech	Transformer	WER (%)	6.1	None	Yes	6.8
61	d'Ascoli et al.	2023	MEG	35	Semantic decod	Self-sup.	Corr.	0.71	Cross-part	No	6.2
62	Wang et al.	2024	EEG	18	Open-vocab	LLM-int.	WER (%)	35.0	None	No	6.4
63	Li et al.	2025	HD-sEMG	3	Phoneme decode	CNN	Acc (%)	82.0	Cross-lang	No	5.9
64	Huth et al.	2022	fMRI	4	Semantic recon.	Transformer	Corr.	0.80	None	No	6.3
65	Ye et al.	2023	ECoG	3	Phoneme speech	LSTM	Acc (%)	74.0	None	No	5.8
66	Defossez et al.	2024	MEG	40	Perceived speech	Foundation M.	Acc (%)	48.0	Cross-part	Yes	6.7
67	Panachakel et al.	2023	EEG	14	Imagined speech	Transformer	Acc (%)	83.0	None	No	6.2
68	Angrick et al.	2025	sEEG	2	Continuous speech	Transformer	WER (%)	9.1	None	No	6.1
69	Akbari et al.	2024	ECoG	1	Attempted speech	Transformer	WER (%)	5.9	None	No	6.4
70	Metzger et al.	2025	Intracort.	1	Attempted speech	LLM-int.	WER (%)	3.8	None	Yes	6.9
71	Defossez et al.	2025	MEG	60	Semantic class	Foundation M.	Acc (%)	52.0	None	Yes	6.8

Column Definitions:

- **ID**: Sequential study identifier (1-71)
- **Authors**: Lead author(s) and year
- **Yr**: Publication year
- **Modality**: Neural recording type (ECoG, Intracort., EEG, MEG, fMRI, sEEG, HD-sEMG, Hybrid)
- **N**: Number of human participants
- **Task Type**: Overt speech, Imagined speech, Attempted speech, Perceived speech, Phoneme classification, Event detection, Semantic reconstruction, etc.

- **Architecture**: Traditional ML, CNN, LSTM, RNN, Seq2Seq, Transformer, Self-supervised, LLM-integrated, Foundation Models
- **Metric**: Primary performance metric (WER, CER, Acc, Corr, WPM, BLEU, ROUGE, MOS, CIL)
- **Performance**: Numerical value of primary metric
- **Generalization**: Cross-participant (Cross-part), Cross-session (Cross-sess), Cross-language (Cross-lang), Long-term stability (Long-term), or None
- **Code**: Code availability (Yes/No)
- **QS**: Quality Score (0-7 scale; ≥ 5.5 = High, 3.5-5.0 = Moderate, ¡3.5 = Low)

Notes on Data Gaps:

- Studies without complete PRISMA database extraction are marked with placeholder values derived from paper text
- Verify N, Task Type, and Architecture against PRISMA extraction register
- Code/Data availability should be confirmed from supplementary materials
- Quality Scores reflect adapted 7-item checklist; verify against individual study assessments